A COCONTINUOUS AND CONTINUOUS FUNCTOR SATISFYING FIVE CONDITIONS INDUCES AN EQUIVALENCE OF TOPOI

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Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The topos morphism associated to a cocontinuous functor. Same chapter, Lemma lemma-cocontinuous-morphism-topoi. Supplies the morphism of topoi whose adjunction maps are shown to be isomorphisms.
- The pullback along a cocontinuous functor and its value on objects. Same chapter, Lemma lemma-when-shriek. Identifies the pullback of a sheaf with the value at the image object.
- The pushforward as a limit over morphisms into an object. Same chapter, the description of the pushforward of a cocontinuous functor as a limit over the category of morphisms from the image of an object. Supplies the description of the composite of pullback and pushforward used in the proof.

Source reference. Original lemma 00XO: sources/stacks-sites/pinned-original-source.tex, source lines 7013–7013. Navigation pointer invoked at proof line 7013; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00XR: sources/stacks-sites/pinned-original-source.tex, source lines 6954–6954. Navigation pointer invoked at proof line 6954; the counted span contains the pointer only and the cited result is not reproduced inside it, so

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Lemma 0.1. *Let $\mathcal{C}$, $\mathcal{D}$ be sites. Let $u : \mathcal{C} \to \mathcal{D}$ be a functor. Assume that*

(1) *u is cocontinuous,*
(2) *u is continuous,*
(3) *given $a, b : U' \to U$ in $\mathcal{C}$ such that $u(a) = u(b)$, then there exists a covering $\{f_i : U'_i \to U'\}$ in $\mathcal{C}$ such that $a \circ f_i = b \circ f_i$,*
(4) *given $U', U \in \mathrm{Ob}(\mathcal{C})$ and a morphism $c : u(U') \to u(U)$ in $\mathcal{D}$ there exists a covering $\{f_i : U'_i \to U'\}$ in $\mathcal{C}$ and morphisms $c_i : U'_i \to U$ such that $u(c_i) = c \circ u(f_i)$, and*
(5) *given $V \in \mathrm{Ob}(\mathcal{D})$ there exists a covering of V in $\mathcal{D}$ of the form $\{u(U_i) \to V\}_{i \in I}$.*

Then the morphism of topoi

$$g : Sh(\mathcal{C}) \longrightarrow Sh(\mathcal{D})$$

associated to the cocontinuous functor u by Lemma 00XO is an equivalence.

Proof. Assume u satisfies properties (1) – (5). We will show that the adjunction mappings

$$\mathcal{G} \longrightarrow g_* g^{-1} \mathcal{G} \quad \text{and} \quad g^{-1} g_* \mathcal{F} \longrightarrow \mathcal{F}$$

are isomorphisms.

Note that Lemma 00XR applies and we have $g^{-1}\mathcal{G}(U) = \mathcal{G}(u(U))$ for any sheaf $\mathcal{G}$ on $\mathcal{D}$. Next, let $\mathcal{F}$ be a sheaf on $\mathcal{C}$, and let V be an object of $\mathcal{D}$. By definition we have $g_*\mathcal{F}(V) = \lim_{u(U) \to V} \mathcal{F}(U)$. Hence

$$g^{-1} g_* \mathcal{F}(U) = \lim_{U', u(U') \to u(U)} \mathcal{F}(U')$$

where the morphisms $\psi : u(U') \to u(U)$ need not be of the form $u(\alpha)$. The category of such pairs (U', ψ) has a final object, namely (U, id), which gives rise to the map from the limit into $\mathcal{F}(U)$. Let $(s_{(U',\psi)})$ be an element of the limit. We want to show that $s_{(U',\psi)}$ is uniquely determined by the value $s_{(U,\mathrm{id})} \in \mathcal{F}(U)$. By property (4) given any (U', ψ) there exists a covering $\{U'_i \to U'\}$ such that the compositions $u(U'_i) \to u(U') \to u(U)$ are of the form $u(c_i)$ for some $c_i : U'_i \to U$ in $\mathcal{C}$. Hence

$$s_{(U',\psi)}|_{U'_i} = c_i^*(s_{(U,\mathrm{id})}).$$

Since $\mathcal{F}$ is a sheaf it follows that indeed $s_{(U',\psi)}$ is determined by $s_{(U,\mathrm{id})}$. This proves uniqueness. For existence, assume given any $s \in \mathcal{F}(U)$, $\psi : u(U') \to u(U)$, $\{f_i : U'_i \to U'\}$ and $c_i : U'_i \to U$ such that $\psi \circ u(f_i) = u(c_i)$ as above. We claim there exists a (unique) element $s_{(U',\psi)} \in \mathcal{F}(U')$ such that

$$s_{(U',\psi)}|_{U'_i} = c_i^*(s).$$

Namely, a priori it is not clear the elements $c_i^*(s)|_{U'_i \times_{U'} U'_j}$ and $c_j^*(s)|_{U'_i \times_{U'} U'_j}$ agree, since the diagram

$$\begin{array}{ccc} U'_i \times_{U'} U'_j & \xrightarrow[\mathrm{pr}_2]{} & U'_j \\ {\scriptstyle \mathrm{pr}_1}\downarrow & & \downarrow{\scriptstyle c_j} \\ U'_i & \xrightarrow{c_i} & U \end{array}$$

need not commute. But condition (3) of the lemma guarantees that there exist coverings $\{f_{ijk} : U'_{ijk} \to U'_i \times_{U'} U'_j\}_{k \in K_{ij}}$ such that $c_i \circ \mathrm{pr}_1 \circ f_{ijk} = c_j \circ \mathrm{pr}_2 \circ f_{ijk}$. Hence

$$f_{ijk}^* \left(c_i^* s|_{U'_i \times_{U'} U'_j} \right) = f_{ijk}^* \left(c_j^* s|_{U'_i \times_{U'} U'_j} \right)$$

Hence $c_i^*(s)|_{U_i' \times_{U'} U_j'} = c_j^*(s)|_{U_i' \times_{U'} U_j'}$ by the sheaf condition for $\mathcal{F}$ and hence the existence of $s_{(U',\psi)}$ also by the sheaf condition for $\mathcal{F}$. The uniqueness guarantees that the collection $(s_{(U',\psi)})$ so obtained is an element of the limit with $s_{(U,\psi)} = s$. This proves that $g^{-1}g_*\mathcal{F} \to \mathcal{F}$ is an isomorphism.

Let $\mathcal{G}$ be a sheaf on $\mathcal{D}$. Let V be an object of $\mathcal{D}$. Then we see that

$$g_*g^{-1}\mathcal{G}(V) = \lim_{U,\psi:u(U)\to V} \mathcal{G}(u(U))$$

By the preceding paragraph we see that the value of the sheaf $g_*g^{-1}\mathcal{G}$ on an object V of the form $V = u(U)$ is equal to $\mathcal{G}(u(U))$. (Formally, this holds because we have $g^{-1}g_*g^{-1} \cong g^{-1}$, and the description of g^{-1} given at the beginning of the proof; informally just by comparing limits here and above.) Hence the adjunction mapping $\mathcal{G} \to g_*g^{-1}\mathcal{G}$ has the property that it is a bijection on sections over any object of the form $u(U)$. Since by axiom (5) there exists a covering of V by objects of the form $u(U)$ we see easily that the adjunction map is an isomorphism. □